



**NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY**

FACULTY OF COMPUTING AND INFORMATICS

DEPARTMENT OF COMPUTER SCIENCE

QUALIFICATION: BACHELOR OF ARTIFICIAL INTELLIGENCE HONOURS	
QUALIFICATION CODE: 08BAIH	LEVEL: 8
COURSE: COMPUTATIONAL GAME THEORY	COURSE CODE: COM810S
DURATION: 5.0 Weeks	MARKS: 100

PAPER REVIEW	
EXAMINER:	Professor José G. Quenum
MODERATOR:	Dr Elizabeth Mutua

INSTRUCTIONS

1. This assignment is to be completed individually.
2. For each submission, a repository should be created on GitHub (<https://github.com/>)
3. The submission date is Friday, June 5 2026, at midnight. Please note that commits after that deadline will not be accepted. Therefore, a submission will be assessed based on the repository's clone at the deadline.
4. There will be a deduction of 5 marks (every two day delay) for students who fail to submit on time.
5. In the case of plagiarism (groups copying from each other or submissions copied from the Internet), all submissions involved will be awarded the mark 0, and each student will receive a warning.

Problem 1[100 points]

In this assignment, students are tasked to review a research paper assigned to them. The list of papers is provided in Table 1, with the full reference under the References section. Each paper is assigned a number, which is then allocated to a student number (in a separate file).

Table 1: Reference

Assigned Number	Citation
1	Adam et al. (2021)
2	McAleer et al. (2021)
3	Lanctot et al. (2017)
4	Tammelin (2014)
5	Tuyls et al. (2006)
6	Bowling et al. (2017)
7	Maiti et al. (2025)
8	Park et al. (2019)
9	Cowden (2012)
10	Wu and Zhang (2026)
11	Xiang et al. (2026)
12	Atay and Trudeau (2026)
13	Jané-Ballarín (2025)
14	Tejada and Álvarez Mozos (2018)
15	Kim et al. (2026)
16	Krishnan et al. (2026)
17	Szolnoki et al. (2022)
18	Li and Huang (2026)
19	Chen and Zhu (2026)
20	Cao and Dang (2025)

Each review should include the following sections:

1. Summary of the paper:

- the problem being addressed;
- the main contribution of the work;
- the experimental/theoretical results;

2. The related work (previous research contributions related to the problem);

3. The limitations of the paper.

Submissions will be assessed following the criteria below:

- The quality and succinctness of the summary. (35%)

- A discussion of relevant (reasonable balance between the minimum and maximum numbers of papers) literature. (35%)
- The overall quality of the critique. (30%)

References

- Adam, L., Horčík, R., Kasl, T., and Kroupa, T. (2021). Double oracle algorithm for computing equilibria in continuous games. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 5071–5078.
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- Jané-Ballarín, M. (2025). The essential coalitions index in games with restricted cooperation. *International Journal of Game Theory*, 54(2):34.
- Kim, J., Fang, F., and Sandholm, T. (2026). Watermarking game-playing agents in perfect-information extensive-form games.
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- Park, S., Martins, N. C., and Shamma, J. S. (2019). From population games to payoff dynamics models: A passivity-based approach. In 2019 IEEE 58th Conference on Decision and Control (CDC), pages 6584–6601.
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- Tammelin, O. (2014). Solving large imperfect information games using CFR+. *arXiv preprint arXiv:1407.5042*.
- Tejada, O. and Álvarez Mozos, M. (2018). Graphs and (levels of) cooperation in games: Two ways how to allocate the surplus. *Mathematical Social Sciences*, 93:114–122.
- Tuyls, K., 't Hoen, P. J., and Vanschoenwinkel, B. (2006). An evolutionary dynamical analysis of multi-agent learning in iterated games. *Autonomous Agents and Multi-Agent Systems*, 12(3):257–289.
- Wu, Y. and Zhang, J. (2026). Discounted stochastic stackelberg games for peer-to-peer energy sharing. *Dynamic Games and Applications*.
- Xiang, H., Wang, Y., and Meng, X. (2026). Non-zero-sum rock–paper–scissors evolutionary game with spatial diffusion: Formation of spatio-temporal patterns and chaos. *Dynamic Games and Applications*.